

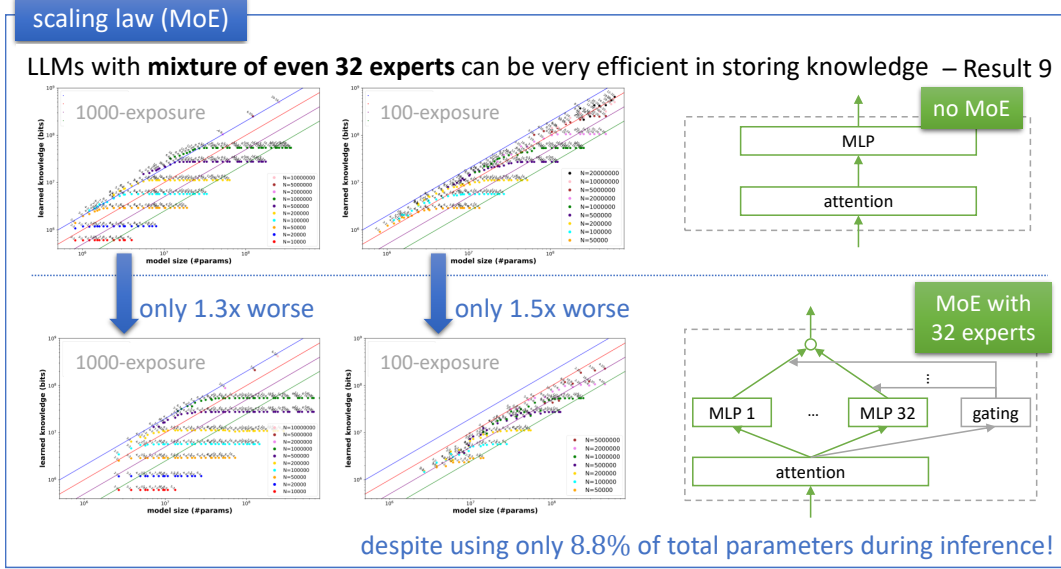


Figure 7: Illustration of Result 9 (details see Figure 16 in Appendix D).

Result 9 (Figure 7). MoE is nearly fully efficient in storing knowledge, capable of leveraging all its parameters despite the sparsity constraint.

Specifically, consider the GPT2-MoE model with 32 experts. If we compute its capacity ratio with respect to the *total* number of parameters and compare that to GPT2:

- in the 1000-exposure settings, the peak capacity ratio decreases by 1.3x; and
- in the 100-exposure settings, the peak capacity ratio decreases by 1.5x.

Remark 9.1 (topk). Result 9 holds even in the “sparsest” setting where $topk = 1$ and $cap_factor = 2$ in the MoE routing. The results are similar when using $topk = 2$ and $cap_factor = 1$ or $topk = 2$ and $cap_factor = 2$ — we discuss more in Appendix D.

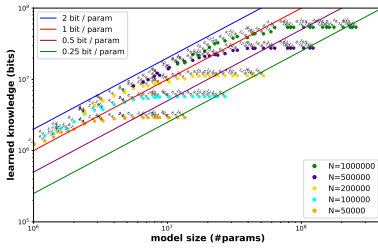
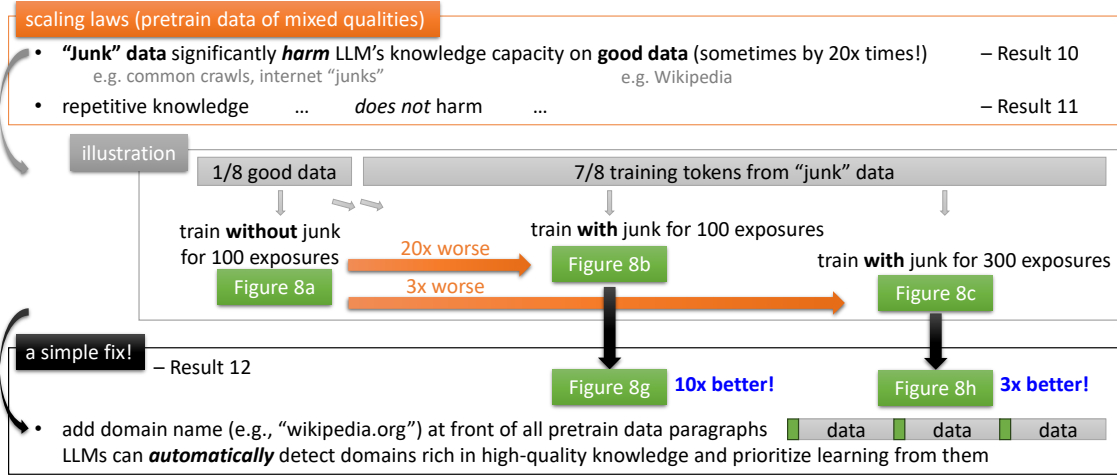
Remark 9.2. It is typically observed in practice that MoE models underperform compared to dense models with the same number of total parameters. We demonstrate that this degradation does not come from the model’s knowledge storage capability.

10 Junk Data vs Scaling Laws

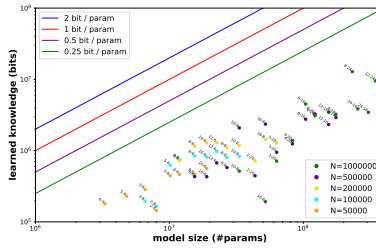
Not all data are useful for knowledge acquisition. For instance, while Wikipedia is full of valuable information, the Common Crawl of web pages may not be (there are also many pieces of information on those webpages, but they may not be useful for a language model to learn, such as the serial number of a random product). How does the presence of low-quality data impact the scaling laws of *useful knowledge capacity*? To investigate this, we create a mixed dataset where:

- 1/8 of tokens originate from $\text{bioS}(N)$ for various N (referred to as *useful data*), and
- 7/8 of tokens originate from $\text{bioS}(N')$ for a large $N' = 100M$ (referred to as *junk data*).

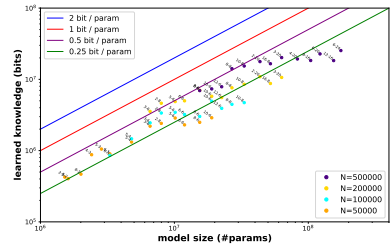
We train models on this mixture, ensuring each piece of useful data is seen for 100 exposures, thus making the total training 8 times longer compared to 100 exposures without junk (i.e., Figure 1(b)). We focus on the capacity ratio of the useful data (the data in $\text{bioS}(N)$) and compare that to



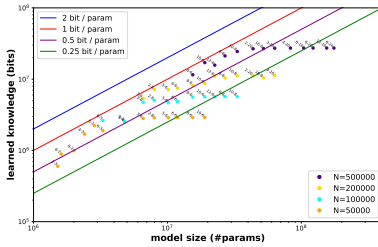
(a) no junk, 100 exposures



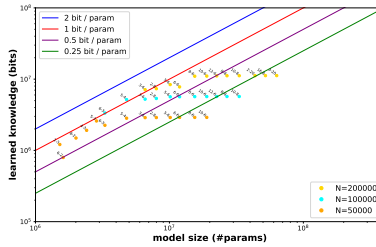
(b) 7/8 junk, 100 exposures



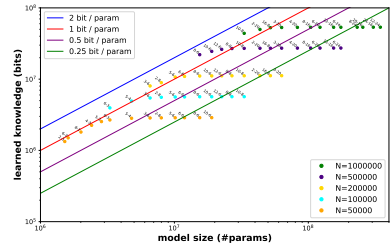
(c) 7/8 junk, 300 exposures



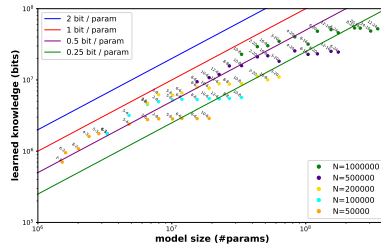
(d) 7/8 junk, 600 exposures



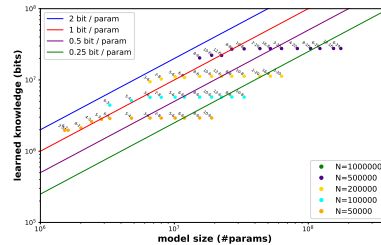
(e) 7/8 junk, 1000 exposures



(f) 7/8 rep-junk, 100 exposures



(g) 7/8 junk, 100 exposures, add special symbol



(h) 7/8 junk, 300 exposures, add special symbol

Figure 8: Capacity ratios with 7/8 junk data (useful data observed 100/300/600/1000 exposures during pretraining).

Conclusions. In Figure 8(b)-8(e), when junk data mimics random knowledge, capacity ratios are *significantly impacted* unless training time is substantially increased. In Figure 8(f), if the junk data is highly repetitive, there is no degradation. In Figure 8(g)+8(h), *adding a special symbol token* to useful data, akin to domain names like wikipedia.org, *mitigates capacity degradation*.

Remarks. Training for Figure 8(e) takes 80x longer than for Figure 8(a), making experiments with larger data sizes N expensive (but anyways unnecessary).

Figure 1(b).²⁰ How much does the capacity ratio degrade in the presence of junk data?

Result 10 (Figure 8(a)-8(e)). When 7/8 of the training tokens come from junk data (i.e., $\text{bioS}(N')$ for $N' = 100M$), transformer’s learning speed for useful data significantly degrades:

- If trained for the same 100 exposures, the capacity ratio may degrade by 20x compared with training without junk (compare Figure 8(b) with Figure 8(a)).
- Even trained for 300/600/1000 exposures, the capacity ratio still degrades by 3x/1.5x/1.3x compared with 100 exposures without junk (Figure 8(c), 8(d), and 8(e) vs. Figure 8(a)).

This underscores the *crucial importance of pretrain data quality*: even if junk data is entirely random, it negatively impacts model’s knowledge capacity even with sufficient training.

In contrast, if 7/8 of data is $\text{bioS}(N')$ with a very small N' , simulating highly repetitive knowledge appearing in training tokens (e.g., “da Vinci painted the Mona Lisa” in millions of variations), this may not affect the model’s capacity for “standard” knowledge (e.g., those with 100 exposures):

Result 11 (Figure 8(f)). If 7/8 of the training tokens come from highly repetitive data (i.e., $\text{bioS}(N')$ for $N' = 1K$), this does not affect the learning speed of useful knowledge:

- The 100-exposure capacity ratio of useful data is unchanged (Figure 8(f) vs. Figure 8(a)).

Finally, if pretrain data’s quality is poor and hard to improve, **a backup strategy exists**:

Result 12 (Figure 8(g)+8(h)). When 7/8 of training tokens are from junk (i.e., $\text{bioS}(N')$ for $N' = 100M$), adding a special token at the start of every useful data greatly improves capacity ratio:

- With 100 exposures, the capacity ratio degrades only by 2x (Figure 8(g) vs. Figure 8(a)).
- With 300 exposures, the capacity ratio matches that of the 100-exposure scaling law without junk (compare Figure 8(h) with Figure 8(a)).

Let us connect Result 12 to practice. First, adding a special token to high-credibility data is very practical: imagine adding the domain name “wikipedia.org” at the beginning of all Wikipedia paragraphs. (Adding a special token to junk data would be less meaningful.)²¹

More generally, one can envision adding domain names (e.g., wikipedia.org) to every piece of the pretraining data. This would significantly enhance the model’s knowledge capacities, because Result 12 demonstrates that **language models can automatically detect which domains are rich in high-quality knowledge and prioritize learning from them**. We emphasize that the model does not need any prior knowledge to identify which domains contain high-quality knowledge; **this process is entirely autonomous**.

11 Conclusion

We investigated the scaling laws of language models, specifically the relationship between model size and the total bits of knowledge stored. Our findings reveal a *precise, universal* scaling law: a sufficiently-trained transformer (i.e., one whose training loss has plateau-ed) can store 2 bits of knowledge per parameter, even when quantized to int8, which is only 1/4 away from the information-theoretical maximum. We also examined how these scaling laws are influenced by various hy-

²⁰The model’s ability to learn from junk data is negligible; each person in $\text{bioS}(N')$ appears only 0.2 times during training when $N = 200k$, or 0.05 times when $N = 50k$.

²¹Result 12 also holds if one adds a (unique) special token for every piece of junk data; however, this could be meaningless as junk data often originates from various websites, making it hard to assign a unique identifier.

perparameters, including training duration, model architectures, floating-point precision, sparsity constraints like MoE, and data signal-noise ratios.

In terms of knowledge capacity, our methodology provides a **more accurate and principled playground** for comparing model architectures, training techniques, and data quality. We believe this playground can assist practitioners in making informed decisions about model selection, training data preparation, and further theoretical research into LLMs. Finally, our research represents an initial step towards addressing a fundamental question: how large does a language model need to be? We hope our findings will inspire further research in this area. Ultimately, we aim to provide a principled answer to the question, “Are language models with 1T parameters sufficient to achieve AGI?” in the future.

APPENDIX

A More on GPT2 Scaling Laws

In this paper, our primary focus is on $\text{bioS}(N)$ for N ranging between 10K and 20M. Notably, $\text{bioS}(20M)$ encompasses approximately 1B bits of knowledge (refer to Theorem 3.2).

GPT2 model. As elaborated in Section 2.3, we refer to the original GPT2 model [26] as GPT2, *after* substituting its positional embedding with *rotary embedding* [7, 31] and removing its dropout layer [33]. These modifications are widely recognized for enhancing performance in language modeling tasks (see also [2] for a controlled experiment comparing that). We explore various GPT2 model sizes, maintaining a dimension-per-head of 64. The notation GPT2- ℓ - h represents the (modified) GPT2 architecture with ℓ layers, h heads, and $64h$ dimensions. The context length is set to 512.

Details on our specifications of LLaMA, Mistral, and other architectures will be provided in Appendix B as needed.

Model sizes. In this study, we calculate model sizes *after excluding* all unused tokens in the embedding layer. For example, while the GPT2 embedding layer typically has $50256 \times (64h)$

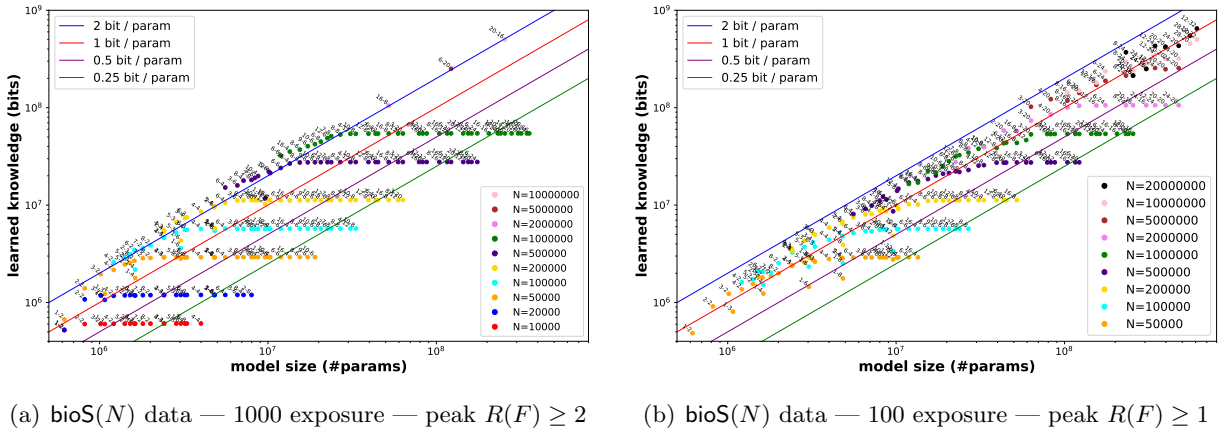


Figure 9: Scaling laws for GPT2 pretrained on $\text{bioS}(N)$ data with fp16 (mixed-precision) for 1000/100 exposures, **now including 1-layer transformers** comparing to Figure 1. **Conclusion:** 1-layer transformers show a minor capacity ratio deficiency, especially in the 100-exposure setting.